

Fubini & Tonelli's Th^m

let $f(x, y)$ be defined in the rectangle

$$I = \{(x, y) : a \leq x \leq b, c \leq y \leq d\}$$

If f is continuous, then

$$\begin{aligned} \iint_I f(x, y) dx dy &= \int_a^b \left[\int_c^d f(x, y) dy \right] dx \\ &= \int_c^d \left[\int_a^b f(x, y) dx \right] dy \end{aligned}$$

Fubini's Th^m: let $I = I_1 \times I_2$, where I_1, I_2 are measurable, and therefore I is measurable. let $f \in L(I)$.

(i.) For a.e. $x \in I_1$, $f(x, y)$ is measurable and therefore integrable on I_2

(ii.) And, as a function of x , $\int_{I_2} f(x, y) dy$ is measurable and therefore integrable on I_1 , and

$$\iint_I f(x, y) dx dy = \int_{I_1} \left[\int_{I_2} f(x, y) dy \right] dx$$

(iii.) Reversing the order

$$\iint_I f(x, y) dx dy = \int_{I_2} \left[\int_{I_1} f(x, y) dx \right] dy$$

Proof: If $f \in L(I)$ and Fubini holds for f , we say it has Property F.

Lemma: A finite linear combination of functions with Property F has Property F.

Proof: Follows from the linearity of Lebesgue Integration.

Lemma: Let $f_1, f_2, \dots$ have Property F. Then if $f_k \nearrow f$ or $f_k \searrow f$ and $f \in L(dx dy)$, then f has Property F.

Proof: Wolog, $f_k \nearrow f$. (Change signs if necessary.) For each f_k , there exists Z_k in $\mathbb{R}$ with $\mu(Z_k) = 0$ s.t. $f_k \in L(dy)$ if $x \notin Z_k$. Let $Z = \bigcup_k Z_k$. Then $\mu(Z) = 0$. If $x \notin Z$, then $f_k(x, y) \in L(dy) \forall k$, and so by Lebesgue Monotone Convergence applied to $\{f_k(x, y)\}$ as functions of y , for $x \notin Z$,
$$h_k(x) = \int f_k(x, y) dy \nearrow h(x) = \int f(x, y) dy.$$

By assumption, $h_k(x) \in L(dx)$, $f_k \in L(dx dy)$, and

$$\iint f_k(x, y) dx dy = \int h_k(x) dx.$$
 Apply LMC again to get the result.

Lemma: If E is a G_δ set, i.e. $E = \bigcap G_k$, and G_1 has finite measure, then χ_E has property $\tilde{F}$.

Proof: W-Z, page 115. //

Lemma: If $\mu(Z) = 0$, then χ_Z has property $\tilde{F}$. Thus, for a.e. $x \in \mathbb{R}$, the set $\{y : (x, y) \in Z\}$ has $\mu = 0$.

Proof: Let H be a G_δ set s.t. $Z \subset H$ and $\mu(H) = 0$. If $H = \bigcap G_k$, $\mu(G_1) = 0$, and so by the previous lemma,

$$\int \left[\int_H \chi_H(x, y) dy \right] dx = \iint \chi_H(x, y) dx dy = 0$$

for a.e. x . If $\{y : (x, y) \in H\}$ has measure zero, then

so does $\{y : (x, y) \in Z\}$ since $Z \subset H$. Then $\chi_Z(x, y)$

is measurable in y and $\int \chi_Z(x, y) dy = 0$. Hence

$\int \left[\int \chi_Z(x, y) dy \right] dx = 0$, which proves the lemma. //

Lemma: Let $E \subset \mathbb{R}^2$. If E is measurable with finite measure, then χ_E has property $\tilde{F}$. //

Proof: We can write $E = H \setminus Z$, where H is G_δ and Z has measure 0.

If $H = \bigcap G_k$, choose G_1 of finite measure. Since

$\chi_E = \chi_H - \chi_Z$, the result follows by the lemmas above. //

The lemmas lead us to the proof of Fubini.

Proof of Fubini: We must show that every $f \in L(dx, dy)$ has property $\mathcal{F}$. Since $f = f^+ - f^-$, we may assume $f \geq 0$.

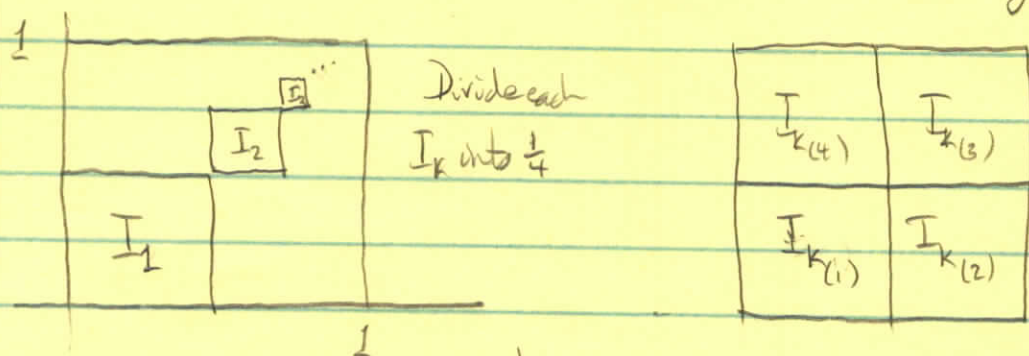
Then, there exist simple measurable f_k with $f_k \nearrow 0$, $f_k \geq 0$.

Each $f_k \in L(dx, dy)$ by the lemma above, we need only show that each has property $\mathcal{F}$. But,

$$f_k = \sum_j \alpha_j^{(k)} \chi_{E_j^{(k)}}, \text{ where } \bigcup_j E_j^{(k)} = E.$$

Since each $E_j^{(k)}$ has finite measures, the result follows by the lemmas above.

The fact that $f \in L(I)$ is needed, as the following example shows.



For each k , let $f = \frac{1}{\mu(I_k)}$ on $I_{k(1)} \cup I_{k(2)}$

$f = \frac{-1}{\mu(I_k)}$ on $I_{k(3)} \cup I_{k(4)}$

let $f \equiv 0$ on $[0,1] \times [0,1] - \bigcup_k I_k$.

Then, $\int_0^1 f(x,y) dx = 0$ for all y , $\int_0^1 f(x,y) dy = 0$ for all x .

However,

$$\iint_I |f(x,y)| dx dy = \sum_k \iint_{I_k} |f(x,y)| dx dy = \sum_k 1 = +\infty.$$

Thus, because $f \notin L(I)$, Fubini does not hold.

Good thing, as $\int_0^1 \left[\int_0^1 f(x,y) dx \right] dy = 0$, but $f \notin L(I)$!

Tonelli's Th^m: let $f(x, y)$ be non-negative and measurable on $I = I_1 \times I_2$, where I_1, I_2 and I are measurable.

Then for a.e. x in I_1 , $f(x, y)$ is measurable and therefore integrable on I_2 . Also, $\int_{I_2} f(x, y) dy$ is measurable and

therefore integrable as a function of x a.e., and

$$\iint_I f(x, y) dx dy = \int_{I_1} \left[\int_{I_2} f(x, y) dy \right] dx.$$

Moreover the process is reversible, and

$$\iint_I f(x, y) dx dy = \int_{I_2} \left[\int_{I_1} f(x, y) dx \right] dy.$$

Ex: let f be non-negative and measurable on E , E measurable in $\mathbb{R}^2$. let

$w(y) = \mu(\{x \in E : f(x) > y, y > 0\})$. Then, by Tonelli

$$\int_E f = \int_0^\infty w(y) dy$$

Fubini does not work on this example.

Applying Fubini & Tonelli

Tonelli - Previous Integral

Fubini - Th^m: let f be non-negative and defined on a measurable set.

Then, if $R(f, E)$ is measurable, then f is measurable.

Proof: Since $R(f, E)$ is measurable, then $\{x \in E: f(x) \geq y\}$

is measurable a.e. y . Therefore, approximating $\{ \}$ by simple

function gives f_k simple, $f_k \nearrow f$. Thus f is measurable.

Defⁿ: convolution $f, g \in L(E)$

$$f * g(x) = \int_{\mathbb{R}} f(x-t) g(t) dt$$

Th^m: (i) $f * g \in L(\mathbb{R})$ with

$$\int_{\mathbb{R}} |f * g| \leq \int_{\mathbb{R}} |f| \cdot \int |g|$$

$$(ii) f * g = g * f$$

Proof of (ii) $f * g = \int_{-\infty}^{\infty} f(x-t) g(t) dt$

$$= \int_{-\infty}^{\infty} f(\tau) g(\tau-x) d\tau = g * f$$

$$\left[\begin{array}{l} \tau = x-t; d\tau = -dt \\ t = \tau-x \\ \int_{-\infty}^{\infty} \rightarrow \int_{+\infty}^{-\infty} \end{array} \right]$$